Asymptotic Covariance of Linear Stochastic Approximation with Markov Noise

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Abstract

For the linear iteration $\theta_{k+1} = \theta_k - \beta_k A\theta_k + \beta_k V_k$ with noise V_k generated by a Markov chain, we prove that $\beta_k^{-1}\mathbb{E}[\theta_k\theta_k^\top]$ converges to the solution Σ of a Lyapunov equation whose right-hand side is the long-run covariance Γ of the noise. The proof reduces Markov noise to martingale-difference noise with the Poisson equation, and then analyses a deterministic recursion for the second moments, as in the i.i.d. case. Only two lemmas involve probability; the rest is matrix analysis. We verify the hypotheses for noise driven by a finite chain and for noise driven by eligibility traces. This is the first of four steps towards the asymptotic covariance of actor-critic algorithms: single time scale with additive noise (this note), two time scales, multiplicative noise, and the transfer to the nonlinear algorithm.

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1 Setting and assumptions

Let $(\Omega, \mathcal{F}, (\mathcal{F}_k)_{k\geq 0}, \mathbb{P})$ be a filtered probability space, $A \in \mathbb{R}^{d\times d}$, and consider

$$\theta_{k+1} = \theta_k - \beta_k A\theta_k + \beta_k V_k, \qquad k \geq 0, \tag{1}$$

with θ_0 $\mathcal{F}_0$-measurable and square integrable, and V_k $\mathcal{F}_{k+1}$-measurable. Norms of matrices are operator norms.

Assumption 1.1 (Step sizes)**.** $\beta_k > 0$ is deterministic, $\beta_k \to 0$, and $\beta_{k+1}^{-1} - \beta_k^{-1} \to \bar{\beta}$ for some $\bar{\beta} \in [0, \infty)$.

For example $\beta_k = \alpha/(k+k_0)$ gives $\bar{\beta} = 1/\alpha$, and $\beta_k = \alpha/(k+k_0)^\rho$ with $\rho < 1$ gives $\bar{\beta} = 0$. In both cases $\sum_k \beta_k = \infty$; this holds in general under Assumption 1.1, since $\beta_k^{-1} \le \beta_0^{-1} + k(\bar{\beta}+1)$ for large k.

Assumption 1.2 (Stability)**.** Every eigenvalue of $A_{\bar{\beta}} := A - \frac{\bar{\beta}}{2} I$ has positive real part.

Assumption 1.3 (Noise)**.** The noise decomposes as

$$V_k = M_{k+1} + h_k - h_{k+1},$$

where h_k is $\mathcal{F}_k$-measurable with $\sup_k \mathbb{E}\|h_k\|^2 < \infty$, M_{k+1} is $\mathcal{F}_{k+1}$-measurable with $\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] = 0$ and $\sup_k \mathbb{E}\|M_{k+1}\|^2 < \infty$, and

$$\Gamma_k := \mathbb{E}[M_{k+1} M_{k+1}^\top] \longrightarrow \Gamma.$$

Section 3 shows that Assumption 1.3 holds for noise generated by a finite Markov chain and by eligibility traces, with Γ the long-run covariance $\sum_{l \in \mathbb{Z}} \mathbb{E}[V_0 V_l^\top]$.

2 Main result

Theorem 2.1. *Under Assumptions 1.1–1.3,*

$$\frac{1}{\beta_k} \mathbb{E}[\theta_k \theta_k^\top] \longrightarrow \Sigma,$$

where Σ is the unique solution of the Lyapunov equation

$$A_{\bar{\beta}} \Sigma + \Sigma A_{\bar{\beta}}^\top = \Gamma. \tag{2}$$

The proof occupies the rest of this section. Throughout, C denotes a constant that does not depend on k and may change from line to line.

2.1 Step sizes

Lemma 2.2. *Under Assumption 1.1, write $\beta_k/\beta_{k+1} = 1 + \beta_k(\bar{\beta} + \epsilon_k)$. Then $\epsilon_k \to 0$, $\beta_{k+1}/\beta_k \to 1$, and $|\beta_{k+1} - \beta_k| \le C\beta_k^2$.*

Proof. The identity $\beta_k/\beta_{k+1} = 1 + \beta_k(\beta_{k+1}^{-1} - \beta_k^{-1})$ shows $\epsilon_k = (\beta_{k+1}^{-1} - \beta_k^{-1}) - \bar{\beta} \to 0$, and then $\beta_k/\beta_{k+1} \to 1$ because $\beta_k \to 0$. Finally $|\beta_k - \beta_{k+1}| = \beta_k \beta_{k+1} |\beta_{k+1}^{-1} - \beta_k^{-1}|$, where the last factor is bounded and $\beta_{k+1} \le C\beta_k$. □

2.2 Removing the telescoping part of the noise

Lemma 2.3. *Let $\theta'_k = \theta_k + \beta_k h_k$. Then*

$$\theta'_{k+1} = (I - \beta_k A)\theta'_k + \beta_k M_{k+1} + \varepsilon_{k+1}, \qquad \varepsilon_{k+1} = \beta_k^2 A h_k + (\beta_{k+1} - \beta_k) h_{k+1},$$

and $\mathbb{E}\|\varepsilon_{k+1}\|^2 \le C\beta_k^4$.

Proof. Substituting the decomposition of V_k in (1),

$$\theta'_{k+1} = \theta_{k+1} + \beta_{k+1} h_{k+1} = \theta_k - \beta_k A \theta_k + \beta_k M_{k+1} + \beta_k h_k + (\beta_{k+1} - \beta_k) h_{k+1}.$$

Now $\theta_k + \beta_k h_k = \theta'_k$ and $-\beta_k A \theta_k = -\beta_k A \theta'_k + \beta_k^2 A h_k$. The bound follows from Lemma 2.2 and $\sup_k \mathbb{E}\|h_k\|^2 < \infty$. □

2.3 The second-moment recursion

Lemma 2.4. *Let $P_k = \mathbb{E}[\theta'_k \theta'^{\top}_k]$. Then*

$$P_{k+1} = (I - \beta_k A) P_k (I - \beta_k A)^{\top} + \beta_k^2 \Gamma_k + R_k, \qquad \|R_k\| \le C\beta_k^2 (\sqrt{\operatorname{tr} P_k} + \beta_k).$$

Proof. Write $\theta'_{k+1} = a + b + c$ with $a = (I - \beta_k A)\theta'_k$, $b = \beta_k M_{k+1}$, $c = \varepsilon_{k+1}$. Since a is $\mathcal{F}_k$-measurable and $\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] = 0$, the tower property gives $\mathbb{E}[ab^{\top}] = 0$. Hence $P_{k+1} = \mathbb{E}[aa^{\top}] + \mathbb{E}[bb^{\top}] + R_k$ with $R_k = \mathbb{E}[ac^{\top} + ca^{\top} + bc^{\top} + cb^{\top} + cc^{\top}]$, and $\mathbb{E}[bb^{\top}] = \beta_k^2 \Gamma_k$. By Cauchy–Schwarz, $\|\mathbb{E}[ac^{\top}]\| \le \sqrt{\mathbb{E}\|a\|^2 \, \mathbb{E}\|c\|^2} \le C\sqrt{\operatorname{tr} P_k}\, \beta_k^2$, $\|\mathbb{E}[bc^{\top}]\| \le C\beta_k \cdot \beta_k^2$ and $\|\mathbb{E}[cc^{\top}]\| \le C\beta_k^4$, using Lemma 2.3. □

2.4 An a priori bound

Lemma 2.5 (Lyapunov matrix)**.** *There is $B \succ 0$ with $A_{\bar\beta}^{\top} B + B A_{\bar\beta} = I$, i.e. $A^{\top} B + B A = I + \bar\beta B$.*

Proof. $-A_{\bar\beta}$ is Hurwitz, so $B = \int_0^{\infty} e^{-A_{\bar\beta}^{\top} t} e^{-A_{\bar\beta} t}\, dt$ converges; it is positive definite, and differentiating $e^{-A_{\bar\beta}^{\top} t} e^{-A_{\bar\beta} t}$ and integrating gives the equation. □

Lemma 2.6. $\sup_k \operatorname{tr}(P_k)/\beta_k < \infty$.

Proof. Let $x_k = \operatorname{tr}(B P_k)$, so that $\lambda_{\min}(B) \operatorname{tr} P_k \le x_k \le \lambda_{\max}(B) \operatorname{tr} P_k$. Using Lemma 2.5,

$$\begin{aligned} \operatorname{tr}\big(B(I - \beta A) P (I - \beta A)^{\top}\big) &= \operatorname{tr}\big(P\,[B - \beta(A^{\top} B + B A) + \beta^2 A^{\top} B A]\big) \\ &= x - \beta \operatorname{tr} P - \beta\bar\beta x + \beta^2 \operatorname{tr}(P A^{\top} B A). \end{aligned}$$

With $a = 1/\lambda_{\max}(B) > 0$ we have $\operatorname{tr} P \ge a x$, so by Lemma 2.4

$$x_{k+1} \le (1 - \beta_k(\bar\beta + a) + C\beta_k^2) x_k + C\beta_k^2 (1 + \sqrt{x_k}).$$

Let $y_k = x_k/\beta_k$ and use $\beta_k/\beta_{k+1} = 1 + \beta_k(\bar\beta + \epsilon_k)$ (Lemma 2.2) and $\beta_k \sqrt{x_k} = \beta_k \sqrt{\beta_k y_k} \le \beta_k (1 + \beta_k y_k)$:

$$\begin{aligned} y_{k+1} &\le \big(1 + \beta_k(\bar\beta + \epsilon_k)\big)\big(1 - \beta_k(\bar\beta + a) + C\beta_k^2\big) y_k + C\beta_k \\ &= \big(1 - \beta_k(a - \epsilon_k + O(\beta_k))\big) y_k + C\beta_k. \end{aligned}$$

Choose k_1 such that for $k \ge k_1$ the coefficient is at most $1 - a\beta_k/2$ and $a\beta_k/2 \le 1$. Then $Y = \max(y_{k_1}, 2C/a)$ satisfies: $y_k \le Y$ implies $y_{k+1} \le (1 - a\beta_k/2) Y + C\beta_k \le Y$. By induction $y_k \le Y$ for $k \ge k_1$. □

2.5 A deterministic lemma

Lemma 2.7. *Let $\mathcal{L}$ be a linear map on a finite-dimensional real vector space whose eigenvalues all have positive real part. Let $\beta_k \to 0$ with $\sum_k \beta_k = \infty$, let $\delta_k \to 0$, and let $e_{k+1} = (I - \beta_k \mathcal{L}) e_k + \beta_k \delta_k$. Then $e_k \to 0$.*

Proof. As in Lemma 2.5 there is an inner product $\langle \cdot, \cdot \rangle_*$ with $\langle \mathcal{L} v, v \rangle_* \ge a \|v\|_*^2$ for some $a > 0$. Then $\|(I - \beta\mathcal{L}) v\|_*^2 = \|v\|_*^2 - 2\beta \langle \mathcal{L} v, v \rangle_* + \beta^2 \|\mathcal{L} v\|_*^2 \le (1 - 2a\beta + C\beta^2) \|v\|_*^2$, so $\|I - \beta_k \mathcal{L}\|_* \le 1 - a\beta_k$ for large k, and

$$\|e_{k+1}\|_* \le (1 - a\beta_k) \|e_k\|_* + \beta_k \|\delta_k\|_*.$$

Fix $\eta > 0$ and k_2 with $\|\delta_k\|_* \le a\eta$ for $k \ge k_2$. Then $u_k = \|e_k\|_* - \eta$ satisfies $u_{k+1} \le (1 - a\beta_k) u_k$ for $k \ge k_2$. If $u_k \le 0$ for some $k \ge k_2$ it stays so; otherwise $u_k \le u_{k_2} \prod_{j=k_2}^{k-1} (1 - a\beta_j) \to 0$ because $\sum \beta_j = \infty$. Hence $\limsup \|e_k\|_* \le \eta$ for every $\eta > 0$. □

2.6 Proof of Theorem 2.1

Let $S_k = P_k/\beta_k$ and $\mathcal{A}S = AS + SA^\top$. By Lemma 2.4,

$$\frac{P_{k+1}}{\beta_k} = S_k - \beta_k \mathcal{A}S_k + \beta_k^2 A S_k A^\top + \beta_k \Gamma_k + \beta_k \frac{R_k}{\beta_k^2}.$$

Multiplying by $\beta_k/\beta_{k+1} = 1 + \beta_k(\bar\beta + \epsilon_k)$,

$$S_{k+1} = S_k - \beta_k(\mathcal{A}S_k - \bar\beta S_k - \Gamma) + \beta_k \delta_k,$$

where

$$\delta_k = \epsilon_k S_k + (\Gamma_k - \Gamma) + \beta_k A S_k A^\top + \frac{R_k}{\beta_k^2} + \beta_k(\bar\beta + \epsilon_k)\Big(- \mathcal{A}S_k + \beta_k A S_k A^\top + \Gamma_k + \frac{R_k}{\beta_k^2}\Big).$$

By Lemma 2.6, S_k is bounded and $\operatorname{tr} P_k = O(\beta_k)$, so $\|R_k\|/\beta_k^2 \le C(\sqrt{\operatorname{tr} P_k} + \beta_k) \to 0$; with $\epsilon_k \to 0$ and $\Gamma_k \to \Gamma$, every term of δ_k tends to zero.

Define $\mathcal{L}S = \mathcal{A}S - \bar\beta S = A_{\bar\beta} S + S A_{\bar\beta}^\top$ on symmetric matrices. Its eigenvalues are the sums $\lambda_i + \lambda_j$ of eigenvalues of $A_{\bar\beta}$, which have positive real part; in particular $\mathcal{L}$ is invertible and (2) has a unique solution Σ. With $e_k = S_k - \Sigma$,

$$e_{k+1} = (I - \beta_k \mathcal{L}) e_k + \beta_k \delta_k,$$

so $e_k \to 0$ by Lemma 2.7 (recall $\sum \beta_k = \infty$).

Finally, $\theta_k = \theta'_k - \beta_k h_k$, so

$$\|\mathbb{E}[\theta_k \theta_k^\top] - P_k\| \le 2\beta_k \sqrt{\mathbb{E}\|h_k\|^2 \operatorname{tr} P_k} + \beta_k^2 \mathbb{E}\|h_k\|^2 = O(\beta_k^{3/2}),$$

and therefore $\beta_k^{-1}\mathbb{E}[\theta_k\theta_k^\top] - S_k \to 0$. □

3 Verifying the noise assumption

3.1 Noise generated by a finite Markov chain

Let (Y_k) be a Markov chain on a finite set $\mathcal{Y}$ with irreducible aperiodic transition matrix Q and stationary law ν, $\mathcal{F}_k = \sigma(\theta_0, Y_0, \dots, Y_k)$, and $V_k = v(Y_k, Y_{k+1})$ with $\sum_{y,y'} \nu(y) Q(y,y') v(y,y') = 0$. Let $\bar v(y) = \sum_{y'} Q(y,y') v(y,y')$, let $Z = (I - Q + \mathbf{1}\nu^\top)^{-1}$ be the fundamental matrix, and $\hat h = Z\bar v$ (applied componentwise).

Lemma 3.1 (Poisson equation)**.** *$\hat h - Q\hat h = \bar v$, and with $M_{k+1} = v(Y_k, Y_{k+1}) - \bar v(Y_k) + \hat h(Y_{k+1}) - (Q\hat h)(Y_k)$ and $h_k = \hat h(Y_k)$ we have $V_k = M_{k+1} + h_k - h_{k+1}$ and $\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] = 0$.*

Proof. $\nu^\top \bar v = 0$ by assumption, so $\mathbf{1}\nu^\top \bar v = 0$ and $(I - Q)Z\bar v = (I - Q + \mathbf{1}\nu^\top)Z\bar v = \bar v$. The decomposition follows from $\bar v = \hat h - Q\hat h$, and $\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] = \bar v(Y_k) - \bar v(Y_k) + (Q\hat h)(Y_k) - (Q\hat h)(Y_k) = 0$ by the Markov property. □

Lemma 3.2 (Long-run covariance)**.** *Let $m(y) = \mathbb{E}[M_{k+1} M_{k+1}^\top \mid Y_k = y]$ (which does not depend on k) and $\Gamma = \sum_y \nu(y) m(y)$. Then $\Gamma_k \to \Gamma$ geometrically fast for every initial law, and*

$$\Gamma = \sum_{l \in \mathbb{Z}} \mathbb{E}_\nu[V_0 V_l^\top],$$

the series converging absolutely.

Proof. $\Gamma_k = \sum_y \mathbb{P}(Y_k = y)m(y)$ and $\mathbb{P}(Y_k = \cdot) \to \nu$ geometrically. For the identity, start the chain from ν and let $T_n = \sum_{k<n} V_k = \sum_{k<n} M_{k+1} + \hat{h}(Y_0) - \hat{h}(Y_n)$. The martingale differences are orthogonal, so $\frac{1}{n}\mathbb{E}[T_n T_n^\top] \to \Gamma$, the boundary terms contributing $O(n^{-1/2})$ by Cauchy–Schwarz. On the other hand $\frac{1}{n}\mathbb{E}[T_n T_n^\top] = \sum_{|l|<n}(1 - |l|/n)\mathbb{E}_\nu[V_0 V_l^\top]$, and for $l \geq 1$, $\mathbb{E}_\nu[V_0 V_l^\top] = \mathbb{E}_\nu[V_0\,(Q^{l-1}\bar{v})(Y_1)^\top]$ with $\|Q^{l-1}\bar{v}\|_\infty \leq C\rho^{l-1}$ for some $\rho < 1$, since $\nu^\top \bar{v} = 0$. So the series converges absolutely, and the Cesàro means converge to its sum. □

Corollary 3.3. *For noise generated by a finite Markov chain, Assumption 1.3 holds and Theorem 2.1 applies with* Γ *the long-run covariance of the noise.*

3.2 Noise generated by eligibility traces

In TD(λ) and in actor-only and actor-critic methods the noise depends on a trace, $f_k = F(Y_k, Y_{k+1})z_k + e(Y_k, Y_{k+1})$ with $z_k = c(Y_k)z_{k-1} + \phi(Y_k)$, where the spectral radius of $Q\,\mathrm{diag}(c)$ is less than one. The pair (Y_k, z_k) is a Markov chain on an infinite state space, but its Poisson equation has an explicit affine solution. Let U and h be as in the closed form for Γ (TD(λ) chapter, Theorem 4.4, and `experiments/trace_cov.py`): $U = \bar{F} + Q\,\mathrm{diag}(c)U$, $w = \bar{e} + Q[U\phi]$, $h = Zw$.

Lemma 3.4. *$H(y, z) = U(y)z + h(y)$ solves the Poisson equation of the chain (Y_k, z_k):*

$$H(y,z) - \mathbb{E}\big[H(Y_1, z_1) \mid Y_0 = y, z_0 = z\big] = \bar{F}(y)z + \bar{e}(y) = \mathbb{E}[f_0 \mid Y_0 = y, z_0 = z].$$

Proof. Since $z_1 = c(Y_1)z + \phi(Y_1)$, the conditional expectation equals $(Q\,\mathrm{diag}(c)U)(y)\,z + (Q[U\phi])(y) + (Qh)(y)$. Subtracting from $H(y, z)$ and using $U - Q\,\mathrm{diag}(c)U = \bar{F}$ and $h - Qh = w - \mathbf{1}\nu^\top w = w$ (because $\nu^\top w = \mathbb{E}f_0 = 0$) leaves $\bar{F}(y)z + w(y) - (Q[U\phi])(y) = \bar{F}(y)z + \bar{e}(y)$. □

Corollary 3.5. *With $h_k = H(Y_k, z_k)$ and $M_{k+1} = f_k - \mathbb{E}[f_k \mid \mathcal{F}_k] + H(Y_{k+1}, z_{k+1}) - \mathbb{E}[H(Y_{k+1}, z_{k+1}) \mid \mathcal{F}_k]$, trace-driven noise satisfies Assumption 1.3 whenever the traces have bounded second moments (for instance $\sup_y c(y) < 1$, or traces reset at restarts), and $\Gamma = \mathbb{E}[M_{k+1}M_{k+1}^\top]$ under stationarity equals the closed-form long-run covariance of the TD(λ) chapter.*

Proof. The decomposition $f_k = M_{k+1} + h_k - h_{k+1}$ and the martingale property follow from Lemma 3.4 exactly as in Lemma 3.1. The identification of Γ follows as in Lemma 3.2, using that the trace forgets its initial value geometrically. □

Remark 3.6 (Numerical checks). (i) For a random 7-state chain, $\mathbb{E}_\nu[MM^\top]$ computed from Lemma 3.1 equals the long-run covariance to relative error 3×10^{-16}; for trace-driven noise, the corresponding quantity from Lemma 3.4 equals the closed form to 7×10^{-16}. (ii) For a 2×2 iteration with Markov noise and $\beta_k = 2/(k + 50)$, a simulation with 20,000 runs gives $\beta_k^{-1}\mathbb{E}[\theta_k\theta_k^\top] \approx \left(\begin{smallmatrix} 0.482 & -0.075 \\ -0.075 & 0.518 \end{smallmatrix}\right)$ at $k = 40{,}000$, against $\Sigma = \left(\begin{smallmatrix} 0.479 & -0.070 \\ -0.070 & 0.523 \end{smallmatrix}\right)$ from (2).

4 What remains

- *Two time scales.* The same reduction to martingale differences applies to each component; the second-moment analysis then follows Konda (2002, Thm. 7.6), with the cross covariance Γ_{12} of the two martingale parts. The goal is the analogue of Thm. 7.8 for Markov noise: the slow component has the covariance of the single-time-scale iteration with noise $V - A_{12}A_{22}^{-1}W$.

- *Multiplicative noise.* In TD and actor-critic the matrix A is itself random, $A(Y_k, z_k)$. Its fluctuation $A(Y_k, z_k) - \bar{A}$ has a Poisson decomposition too, and contributes terms of order $\beta_k \, \mathbb{E}\|\theta_k\|^2 = O(\beta_k^2)$ to the recursion, once an a priori bound like Lemma 2.6 is established.
- *Nonlinear algorithms.* The transfer from the linearized iteration to the actual actor-critic algorithm is left as an explicit hypothesis.

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